MIL-G-13507(Ord)
24 June 1954
SUPERSEDING
Army 5I-78-1OO
6 June 1949

MILITARY SPECIFICATION
GEAR, OIL, M6A1E1

1. SCOPE

1.1 This *specification covers an* oil gear, which is a component part *of the System, Local Control,* M16A1E1 for use with Gus, Twin, 40mm, Self-Propelled,. M42.

2. " APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS, AND PUBLICATIONS "

2.1 *The* following specifications and drawings of the issue In effect on the date *of invitation* for bids, form a part of this specification:

SPECIFICATIONS
Military

MIL-G-3278	Grease; Aircraft and Instruments (for Low and High Temperatures).
MIL-O-6083	Oil; Preservative, Hydraulic Equipment
MInL-s-l0379	Suppression, Radio Interference, GeneraI Requirements for Vehicles (and Vehicular Sub-Assemblies).
MIL-U-12114	Unit, Hydraulic Transmission F7666857, D7692321, and D75082.
MIL-M-12652	*Motor, D-C, 24* Volts, 1-1/2 H. P.
U. S. Army 51-7	Instruments, Fire Control General Specification Governing the Manufacture, and Inspection of

DRAWINGS
Ordnance Corps

F7665297	Gear, Oil, M6A1E1

(copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer).

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3. REQUIREMENTS

3.1 General.- Unless otherwise specified, the requirements common to all fire control instruments as contained in Specification 51-7 shall be mandatory.

3.2 Material - Material shall be in accordance with the applicable drawings and specifications forming a part of this specification.

3.2.1 Grease.- Grease to be used for bearings (except those in the synchro unit) and gears, shall be in accordance with Specification MIL-G-3278.

3.2.2 Hydraulic oil.- Hydraulic oil shall be in accordance with Specification MIL-O-6083. Clean oil shall be used exclusively

9.3 Electrical

3.3.1 Power input. Each oil gear shall meet the performance requirements *of* this specification when 24volts d-c is applied to the motor and 115 volts, 60 cycles, " is applied to the synchros and relays. Two hours operation at 130 volts a-o shall not cause damage.

3.3.2 High potential. - There shall be no puncture of insulation, no sparkover, or decrease in dielectric strength of the insulation sufficient to cause failure *of* the component under test when it is subjected to the test specified in 4.2.1 .

3.3.3 Insulation resistance.- The insulation resistance shall be not less than 5 megohms when subjected to the test specified in 4.2.2.

9.3.4 Radio interference suppression. - The oil gear shall meet the requirements of Specification M I L = S 1 0 3 7 9 .

3.4 Performance

3.4.1 Transmission and valve assembly .- Transmission and valve aSSeMby shall meet the requirements of the following Sub-paragraphs when tested in accordance with 4.9.2.

3.4.1.1 Replenishing pump Pressure. - With control at neutral stroke, pressure shall be beween 150 and 160 pounds per square inch.

3. 4.1.2 Dither.- The dither mechanism shall vibrate to a degree perceptible to the eye and touch.

9.4.1.3 Differential synchro. - When the differential is electrical zero connected, the hydraulic motor speed shall not exceed 60 rpm.

3.4,1.4 Hydraulic motor output speed. - When the synchros are deenergized and the unit is accelerated to full speed by displacing the differential synohro shaft to the maximum extent, the hydraulic motor speed shall not be less than 3000 rpm in both directions. The variation in the readings for each direction of rotation shall not exceed 150 rpm.

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9,4.1. S Relief valve check - When the dial of the transmitter gearing is rotated by hand, the output torque, measured at *the hydraulic* motor shaft, shall be between 50 and 75 pound-inches.

3.4.1.6 Surge.- At constant rate, the change of rate shall not exceed 1-1/4 mils of gun motion when measured in accordance with 4*3.2.5-

3.4.2 Complete oil gear. - Complete Oil gear shall meet the requirements *of the* following sub-paragraphs when tested in accordance with 4,3.3.

3.4.2,1 Oil temperature

3.4.2.161 Starting ambient temperature between 65" and 75". - The maximum temperature of the oil shall not exceed 200°F after 2 hours operation with zero output rate.

3.4.2.1.2 Starting ambient temperature of 125°F. - The oil gear shall not be damaged and shall meet the requirements of this specification when tested after being operated for a period of one hour with zero output rate for 95 percent of the time.

3.4.2.2 Replenishing pump pressure. - With control at neutral stroke, oil pressure at filter shall be between 150 and 160 pounds per square inch.

3.4.2. S Power input. - When the stroke control is set at neutral, the power Input to the electric *drive* motor shall not exceed 950 watts when the 0il temperature is 100°F, 900 watts when the oil temperature is 140°F. The thermometer shall be between 4 and 6 inches from the bottom of the case.

3.4.2.4 Output speed. - When the oil gear is connected to a load of 94 inch-pounds and accelerated to full speed, the coupling of the output coupling shaft shall rotate at a minimum speed of 500 rpm.

3.4.2.5 Creepage. - When the synchro excitation is switched off, the output coupling shaft of the oil gear shall not turn more than 50 rpm in either direction.

3.4.2.6 Limit cam settings. - The limit cams shall be set at -3.0 degrees ± 0.5 in depression and 85 degrees ± 1. O in elevation. When the limit cam is set correctly in the elevation position and the oil gear is slewed into each limit at a speed *of* 25 degrees per second or greater the control mechanism shall return to its neutral position.

3,4.2.7 Clutches.- The slewing clutch and boresight clutch shall operate smoothly and both clutches shall engage and disengage positively when the levers are operated in the appropriate manner. Minimum slewing clutch engagement shall be 3/32 inch.

3.4. 2. 8 Surge. - At constant rate the change of rate shall not exceed 1-1/4 roils of gun motion when measured in accordance with 4.3.3.4.

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S. 4.3 Motor. - Motor shall be in accordanoe with Specification *MIL-M-12652.*

3.4.4 Hydraulic transmission unit. - Hydraulic transmission unit shall be in accordance with specification MIL-U-12114.

S. 5 Environmental

3.5. 1 Temperature. - The oil gear shall meet the eleotrical and *performance* requirements of this specification within an air temperature range extending from plus 51. 5°C (+125°F) (minimum exposure *of* 4 hours with full impact of solar *radiation,* 360 BTU/Ft. Sq/Hr.), to minus 31. 2°C (- 25°F) (minimum exposure of 3 days without benefit of solar radiation). The oil gear shall be capable of safe storage within the air temperature range extending from plus 71°C (+ 160°F) (for periods as long as 4 hours per day) to minus 62.2* (- 80°F) (for periods of at Ieast 3 days duration). Teat shall be conducted in accordance with 4.4.2.

3.5.2 Humidity The oil gear shall meet the electrical and performance requirements of this specification immediately after being subjected to the humidity test specified in 4.4.3.

3. 5. 3 Rain. - There shall be no evidence of water penetration or damage to the oil gear after it be been subjected to the rain test in accordace with 4.4.4.

3.5.4 Vibration. - The oil gear shall not *suffer damage or* fail to meet the electrical and performance requirement of this specification immediately after, being subjected to the vibration test specified in 4.4. S.

3. S. 5 Shock- The oil gear shall not suffer damage or fail to meet the electrical and performance requirement of this specification immediately after being subjected to the shook test specified in 4.4.6.

3.6 Workmanship. Workmanship shall be in accordance with Specification 61-7.

3.6.1 sealing. - The oil gear shall not leak Oil.

3.6.2 Cleanliness.- There shall be no evidence of dust, dirt, lint, or other foreign matter in the oil which will affect proper operation of the oil gear.

4. SAMPLING , INSPECTION, AND TEST PROCEDURES

4.1 General.- Unless otherwise specified, each oil gear shall be inspected for conformance with this specification and Specification 51-7. Unless otherwise specified, standard *ambient tests shall be performed* at a temperature *of* 60°F to 90°F at a barometric pressure of 28 to 32 inches of mercury and a relative humidity not greater than 80 percent.

4.2 Electrical

4.2. 1 High POtential:- Electrical component of the oil gear shall be subjected to a potential of 900 volts, with a frequency of 60 cycles per second *for* a period

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of one minute. Three of the first five and one out of every additional 50 or fraction thereof shall be subjected to the test voltage which shall be applied at a rate not exceeding 500 volts (rms) *per second.* For all other units the test voltage, shall be applied instantaneously for a period of not less than 5 seconds or more than 30 seconds. The motor capacitor and starting relay shall be disconnected to prevent damage. Insulation resistance test shall be performed immediately after this test.

4.2.2 Insulation resistance. - The insulation resistance shall be measured by an approved method using a source *of* d-o potential of 500 volts. The capacitor shall be disconnected for this test.

4.2.3 Radio interference. The oil gear shall be tested for radio interference in accordance with Specification MIL-S-10379.

4.3 Performance

4.3.1 General.- Tests shall be conducted when hydraulic oil is at a temperature of 115°F plus or minus 4°F.

4.9.2 Transmission and valve assembly .- All output values shall apply to both clockwise and counterclockwise rotation.

4.& 2.1 Replenishing pump pressure. - With assembly in place and power motor energized, pressure shall meet the *requirements of 3.4.* 1. 1.

4.3.2.2 PilOt piSton p0sitiOning• The position of pilot piston shall be tested as follows:

a. *Adjust* synchro so that stops are equidistant from the pin on the differential stator and the differential damping spring is in approximate horizontal position.

b. Place unit in test stand.

c. Rough adjust for equal speed in both directions.

d. Electrically zero and adjust pilot piston for approximate zero output of flywheel (60 rpm maximum *of transmission output shaft).*

e. Reenergize synchro and adjust motion *of* flywheel to approximate zero (60 rpm maximum *of* transmission output shaft) by adjusting mechanical zero *lever.*

f. Adjust differential synchro with electrical zero excitation@ approximate zero output of flywheel (60 *rpm maximum of* transmission output shaft).

4.3.2.3 Differential synchro. - The differential synchros shall be switched to electrical zero *connections and* output speed shall be observed and recorded. If the hydraulic motor speed is greater than 60 rpm, loosen clamp screws and rotate stator as required.

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4. 3. 2.4 Relief valve check. - The torque drum shall be engaged with the output shaft. The maximum output torque in each direction shall be measured to check the transmission *relief valve. The* dial of the transmitter gearing shalI be rotated by hand until maximum pull on the spring balance is reached for output rotation in both directions. This test should not *be* applied for more than 5 seconds because of the excessive heating in the high pressure relief valve. The torque developed shall be between 50 and 75 pound-inches at the hydraulic motor shaft.

4.3.2.5 Surge. - Change of rate shall be measured by matching the rate from a constant speed mo against the rate of the transmission and valve assembly. These rates are the inputs to a differential; the output *of the* differential will the be the change in rate, Rates shall be *measured between* 200 and 500 *mils* per second.

4.3.3 Complete oil gear. - During tests, limit cam shall be set for azimuth operation.

4.3.3.1 Replenishing Pump pressure. - Power shall be applied at motor and cheek shall be made for proper rotation by observing oil pressure at filter. Pressure shall be in accordance with 3.4.2.2. Motor rotation should be correct when the plus and minus terminals of the d-c power source are connected to motor leads M and G, respectively.

4.3.3.2 outPut speed. - The limit stop screw shalI be adjusted in clockwise and coounterclockwise direction until output speed meets the requirements of 3.4.2.4.

4.3.3.3 *Creepage.*- With the oil gear having been synchronized previously with the test director and at rest for at least 5 Seconds, the creepage shall be checked by switching off the synchro excitation. The unit shall meet the requirements of 3.4.2.7,

4.3.9.4 Surge. - Change *of* rate shall be measured by matching the rate from a constant speed motor against the rate of the oil gear *assembly.* These rates are the Inputs to a differential. The output of the differential will then be the change in rate. Rates shall be measured between zoo and 500 mils per second.

4.4 Environmental

4.4.1 General. The environmental tests specified shall be limited to those contained in the following paragraphs. Three of the first five oil gears and one out of every additional 100 or fraction thereof shall be tested.

4.4.2 Temperature. - The oil gear shall be exposed to an air temperature of minus 62. 2°C (-80°F) for a period *of 3 days.* The air temperature then shall be raised to minus 31.2°C (-25°F) and after the internal temperature of the oil gear has reached equilibrium (approximately 4 hours), it then shall be operated until the oil temperature has stabilized. The oil gear shall then be tested for conformance with the performance requirements of this specification. The air temperature shall be raised to plus 71°C (± 160°F) and after the internal temperature of the oil gear has reached equilibrium at that temperature it shall be allowed to remain at that temperature for a period of 4 hours. The air temperature shall be reduced to plus 51. 5°C

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(+125°F) and after the internal temperature of the oil gear has reached equilibrium at that temperature, the top surface shall be exposed to 360 BTU/Ft. Sq/Hr., for a period of 4 hours. The oil gear then shall be tested for conformance with the performance requirements of this specification. The unit shall be allowed to return to normal room temperature and test shall be made for conformance *with 3.5.1.*

4.4.3 Humidity - A test chamber that is capable of being sealed shall be used: the source of heat shall be arranged in such a manner that the radiant heat will not fall upon the unit being tested. An oil gear shall be placed in the chamber, the temperature of which is adjusted to plus 46. 1°C ± 1.1°C (± 115°F ±2°F) and the relative humidity adjusted to 95 percent* 5 percent. The unit shall remain under the above conditions for a period of 6 hours. The temperature* of the chamber shall be readjusted to plus 26. 7°C ± 1. 1°C (± 80°F ±2°F) and the oil gear shall be allowed to remain under this new condition for a period of 6 hours. The transitional period shall be 2 hours; the rate of change during that period shall be uniform. The above cycle shall be repeated 10 times. Immediately after this test, the oil gear shall meet the requirements *of* 3.5.2.

4.4.4. Rain. - The oil gear shall be mounted in the test chamber to simulate installation conditions. The rain test chamber temperature shall be maintained between 20°C to 30°C *(68°F to 86°F)* throughout the test period. A simulated rainfall of *4* plus 1 inch per hour shall be produced by a suitable means of such design that the water is emitted in the form of small droplets rather than a fine mist. The temperature of the water shall be maintained between ll°C to 24°C (51. 8°F to 75.2°F). The water spray shall be directed on the oil gear for a period of 2 hours in directions ranging from vertical to 45° from the vertical. The simulated rain shall be directed on all surfaces of the oil gear that are normally subjected to rain when the instrument is in its *operating* position.

4.4.5 Vibration. - The oil gear shall be mounted using its own mounting provisions. to simulate field use. Each unit tested shall be vibrated successively along each of the three mutually perpendicular directions with accelerations and excursions as shown below. Applied frequency shall be changed at a uniform rate between the limits indicated; the rate of change shall not exceed one cycle per second.

Frequency limits	Excursion	Acceleration	Time (each direction)
0- 10 Cps	0.060 inch	-00000000	1 hour
10- 20 Cps	0.030 inch	00-09--00-	2 hours
20-400 Cps		2.5g's	2 hours

4.4. $ Shock. - The oil gear shall be mounted, using its own mounting provisions, to simulate field use. A shock of 20 g's shall be imparted three times in each direction along each of the three mutually perpendicular directions, making a total *of .. 18* shocks. Each shock impulse shall have a time duration of 11 ± 1 milliseconds; the maximum g value shall occur at approximately 5.5 miilliseconds and shall be within * 10 percent. Filters used shall have cut-off frequencies selected to eliminate all harmonics above the fundamental from the measured chock pulse.

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5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. Packaging, packing and marking shall be in accordance with 9instructions forming a PART OF THE CONTRACT OR FURNISHED THE contracting officer in connection with the Contract.

6. NOTES

6.1 Ordering data. - Contracts and invitation *for* bids should *specify the title, number, date* of this specification and type of packing.

NOTICE: This specification, together with speolfications and drawings pertaining to it and bearing a "NOTICE" of similar restrictions, is intended for use only in connection with procurement by the United States Government and shall not be reproduced either wholly or in part except when authorized in connection with Government procurement, nor be used for any other purpose except when specifically authorized by the Chief of Ordnance.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

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